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Baddeleyite Pb—Pb geochronology and paleomagnetic poles for ~1.89–1.86 Ga mafic intrusions from the Dharwar craton, India, and their paleogeographic implications

V. Parashuramulu, Ravi Shankar, D. Srinivasa Sarma *, E. Nagaraju, N. Ramesh Babu

CSIR-National Geophysical Research Institute, Council of Scientific and Industrial Research, Uppal Road, Hyderabad 500 007, India

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ABSTRACT

We present new key paleomagnetic pole at 13°S, 152°E (k = 21, A_{95} = 7.8°) for recently identified 1864.4 ± 2.7 Ma (weighted mean age of four Pb—Pb ages) mafic magmatic event, based on a detailed paleomagnetic study of dolerite dykes and sills intruding Archean basement rocks and Tadipatri formation of the Cuddapah basin, Dharwar craton respectively. The Pb—Pb baddeleyite geochronology yields a crystallisation age of 1867.1 ± 1.0 Ma (MSWD = 1.02) for N77°E trending dyke in the southern region to Cuddapah basin. This new age obtained, confirms the presence of ~1864 Ma magmatic episode with a spatial extent of ~400 km in the Eastern Dharwar craton, within the brief period of ~5 Ma. The paleomagnetic results in these dykes revealed reverse polarity magnetisation direction with mean D = 107°, *I* = 24° (*N* = 13 sites, α_{95} = 10°). Here, we also update the normal polarity magnetic directions on ~1.89 Ga swarm, and the corresponding paleopole situated at 21°N, 336°E (*N* = 79 sites, A_{95} = 3.6°). The paleoposition of India is constrained around the equator during ~1.89–1.86 Ga time. The paleogeographic reconstructions were also been attempted at ~1.89 Ga and ~ 1.86 Ga with available key poles from other cratons, indicates the possibility of single plume acting as a source for two distinguishable radial emplacement of mafic dyke swarms across India (Dharwar and Bastar craton) and Western Australia (Yilgarn craton) within a time span of ~35 Ma. The individual movement of India, Baltica and Siberia with a drift rate of ~5.55 cm/yr towards the south, whereas Amazonia craton has moved rapidly to the north (~24.9 cm/yr), do not suggest the amalgamation of a supercontinent (Columbia/ Nuna) during ~1.88–1.86 Ga time.

1. Introduction

Precambrian Large Igneous Provinces (LIPs) are often associated with significant regional scale uplift, continental rifting and breakup, climatic events and metallogenic systems (Ernst et al., 2005; Ernst and Bleeker, 2010; Ernst and Jowitt, 2013; Ernst and Youbi, 2017). Although such LIPs are extensively eroded during Earth's evolution, their reminiscence is manifested by the extensive networks of mafic intrusions such as mafic dykes and sills. These represent deep-level plumbing systems for magma transportation from source to the surface (Halls, 1982). Mafic dyke/sill swarms are ideal subjects of study due to their ability to retain high-quality U-Pb/ Pb—Pb geochronology, paleomagnetic, isotopic and geochemical characteristics (Halls, 1982), which are useful in constraining the timing of mantle plume activity, and understanding the formation and breakup of ancient supercratons/supercontinents (Halls, 1982; Ernst and Buchan, 1997; Li and Zhong, 2009; Kumar et al., 2017; Meert and Santosh, 2017; Shankar et al., 2018).

The ~1885 Ma mafic-ultramafic magmatic activity is considered as one of the most massive igneous episodes in the Earth's history, with considerable spread over several widely separated continents (French et al., 2008; Minifie, 2010, references therein). This magmatic activity is recorded by mafic intrusions and volcanics in the Superior craton (Heaman et al., 1986; Zhai et al., 1994; Halls and Heaman, 2000; Heaman et al., 2009), the Slave craton (Davis and Bleeker, 2007; Buchan et al., 2016), the Yilgarn craton (Stark et al., 2017; Liu et al., 2018), the Kaapvaal craton (Hanson et al., 2004; Olsson et al., 2016), the Zimbabwe craton (Söderlund et al., 2010; Hanson et al., 2011), the Baltica shield (Klein et al., 2016, references therein), the Amazonia craton (Antonio et al., 2017; Teixeira and Hamilton, 2019), the Dharwar and Bastar cratons (French et al., 2008; Meert et al., 2011; Shellnutt et al., 2018; Sarma, 2020). However, this mafic magmatism is significant not only because of its widespread geospatial extent but also due to its

* Corresponding author.
E-mail address: dssarma@ngri.res.in (D.S. Sarma).

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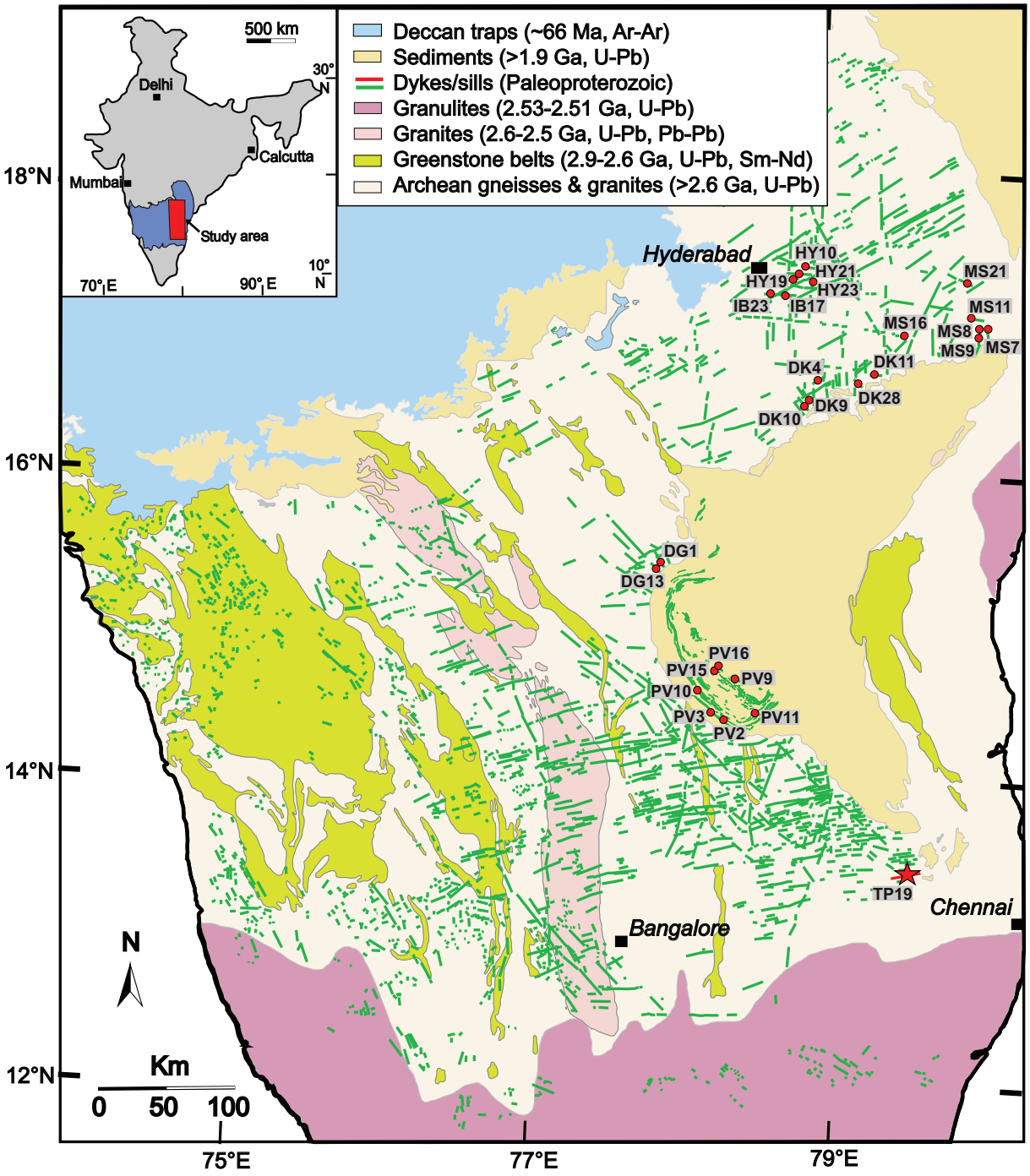


Fig. 1. Simplified geological map of the Dharwar craton showing the distribution of mafic dykes and locations of sampled sites. Red circles represent sites of paleomagnetic studies in the present study and Pb—Pb baddeleyite geochronological studies was performed on site represented by red star. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

concurrence with substantial phases of continental crust formation as well environmental/climatic disturbance along with oceanic anoxic conditions (evidenced by extensive black shale deposits). Furthermore, this event also holds economic interest due to its association with significant Ni—Cu and Platinum Group Element deposits (Barnes et al., 1982; Lesher and Keays, 2002; Naldrett, 1999; Clark and Wares, 2004). The contemporaneous age (~1.88 Ga) of Iron formations has been identified globally as a consequence of significant igneous activity and rapid crustal growth at that time (Rasmussen et al., 2012 and references therein).

The Dharwar and Bastar cratons have witnessed widespread mafic magmatism during 1890–1860 Ma as two significant events. The first event is 1894–1883 Ma LIP (with a weighted mean age of ~1886 Ma, Sarma et al., 2020, references therein), spread across Dharwar and Bastar cratons as mafic dykes and sills, which was first identified by French et al. (2008). About ~20 Ma later, Dharwar craton experienced another mafic magmatic event at ~1864 Ma (Sarma, 2020) in the form of mafic dykes and sills. There is another slightly younger event reported from Bastar craton at ~1850 Ma (Shellnutt et al., 2019). However, several paleomagnetic studies (Radhakrishna et al., 2013b; Belica et al., 2014, references therein) have been reported on these (1885–1865 Ma) dykes/ sills, widely distributed across the craton. These studies revealed

dual polarity remanence directions and interpreted it to be associated with ~1885 Ma LIP. Such interpretation needs to be assessed again in light of recently reported ~1864 Ma event (Sarma, 2020), as the reverse paleomagnetic directions may have acquired during its emplacement.

Here, we present precise Pb—Pb baddeleyite age together with consistent paleomagnetic results on several mafic dykes and sills intruding Archean basement and Proterozoic sedimentary rocks to determine the critical key paleopole at ~1864 Ma and also upgrade the paleopole at ~1886 Ma of Dharwar craton. We attempt paleogeographic reconstruction with obtained key poles and available coeval paleopoles from other cratons to understand the paleoposition of India and its relationship with other cratonic/continental blocks during that time.

2. Background Geology

The Indian subcontinent encompasses several well preserved Archean cratons, includes Singhbhum Craton in the eastern region; Bundhelkhand Craton in the central part, Bastar Craton in the south-central region and the Dharwar Craton in the southern region (Meert et al., 2010; Meert and Pandit, 2015). Eastern Ghats Mobile Belt (EGMB), Southern Granulite Terrane (SGT), extensive basaltic flows of the Deccan traps, Godavari graben, and the Arabia sea demarcate the eastern, southern, north-western, north-eastern, and western boundaries of the Dharwar craton, respectively (Naqvi and Rogers, 1987a, 1987b). Several Proterozoic sedimentary basins were evolved throughout the craton; typical crescent-shaped Cuddapah basin is a major intracratonic basin among them and situated in the south-eastern part of Dharwar craton. The grade of metamorphism varies along with the titled crustal province towards the south (from the north) and show the high grade of southern granulite terrain in the south.

Further, the Dharwar craton had been divided as Western Dharwar Craton (WDC) and Eastern Dharwar Craton (EDC), separated by an N-NW trending fault zone. The WDC and EDC have amalgamated at ~2.5 Ga (Chadwick et al., 2000; Balakrishnan et al., 1990). The WDC comprises of TTG-type peninsular gneisses which accreted in several stages during 3.36 Ga and 3.2 Ga (Meen et al., 1992; Peucat et al., 1993), two phases of greenstone belts, the older Sargur Group, younger Dharwar Supergroup and calc-alkaline to potassic plutons (Chadwick et al., 2000; Jayananda et al., 2006). The EDC comprises of distinctly younger (2.7–2.55 Ga) rocks, includes TTG gneisses and migmatites with limited 3.38–3.0 Ga crust remnants (Balakrishnan et al., 1999; Nutman et al., 1996; Friend and Nutman, 1991; Jayananda et al., 2000; Chardon et al., 2002, 2011) and multiple exposures of 2.7–2.55 Ga greenstone belts (Balakrishnan et al., 1990, 1999; Nutman et al., 1996; Rogers et al., 2007; Sarma et al., 2008, 2011, 2012; Mohan et al., 2013, 2014; Krapež et al., 2019).

The Dharwar craton has experienced episodic emplacement of mafic dyke swarms in at least eight different mafic magmatic events, during the Paleoproterozoic between ~2367 Ma to ~1864 Ma (Sarma, 2020, and references therein). The NW-SE to NE-SW trending giant radiating dyke swarm with an emplacement age of 2367.4 ± 0.73 Ma, is the oldest and most dominant mafic dyke swarm in the Dharwar craton, and it covers most of the EDC (Halls et al., 2007; French and Heaman, 2010; Kumar et al., 2012a; Pivarunas et al., 2018; Söderlund et al., 2019; Sarma, 2020). Two N-S trending dyke swarms were identified with emplacement ages of 2253.0 ± 1.7 Ma (Nagaraju et al., 2018b; Söderlund et al., 2019) and 2215.6 ± 1.5 Ma (French and Heaman, 2010; Srivastava et al., 2011; Kumar et al., 2014; Nagaraju et al., 2018a). Three radiating dyke swarms were recognised with emplacement ages of 2210.4 ± 0.6 Ma (French and Heaman, 2010; Nagaraju et al., 2018b; Yadav et al., 2020; Söderlund et al., 2019; Sarma, 2020), 2180.6 ± 1.7 Ma (French and Heaman, 2010), and 2081.6 ± 0.4 Ma (Kumar et al., 2015, references therin). In addition to these, a Large Igneous Province with ages of 1894–1883 Ma, consisting of mafic dykes in the Bastar craton and mafic-ultramafic sills in the Cuddapah basin was reported by French et al. (2008). A new mafic magmatic event at 1863.8 ± 0.9 Ma (Sarma, 2020), was reported recently for mafic sills of Cuddapah basin and N-S dyke at its northern margin. Additionally, two individual dykes were reported with an emplacement age of 1794 ± 7 Ma (from EDC; Söderlund et al., 2019) and 1957.3 ± 0.7 Ma (from WDC; Yadav et al., 2020). Primary paleomagnetic and anisotropy of magnetic susceptibility results have been reported on many dyke swarms in recent studies (Kumar et al., 2012a, 2012b; Belica et al., 2014; Kumar et al., 2015; Nagaraju et al., 2018a, 2018b; Babu et al., 2018, 2020; Nagaraju and Parashuramulu, 2019; Fig. 1).

Formation of Paleoproterozoic sedimentary basins which overlie the Archean basement seems to be closely related to the emplacement of these dykes (Kumar et al., 2015). The Cuddapah basin, with an aerial extent of ~44,500 km^2 and sediment thickness of 6–12 km (Nagaraja Rao et al., 1987; Ramam and Murty, 1997; Singh and Mishra, 2002) hosts subaerial extrusive and intrusive mafic/ultramafic rocks with emplacement ages of 1885–1864 Ma (U-Pb/Pb-Pb baddeleyite age, French et al., 2008; Sarma, 2020) within its lower stratigraphic horizons.

3. Sampling strategy

Sampling for this study is restricted to fresh insitu outcrops like quarries, road and canal cuts. The oriented block samples (at least five samples per site) for paleomagnetic investigations, from mafic dykes intruding Archean basement rocks of EDC, and mafic sills intruding Tadipatri sedimentary formation of Cuddapah basin, is done with the help of magnetic and sun compasses. A total of 130 samples were collected from 26 sites representing 24 dykes, and 36 sill samples at 7 locations (Fig. 1). These intrusions varied in trend (Fig. 1) and thickness along the dyke and among the dykes. All the samples (dykes and sills) are in dolerite composition. Dykes show vertical dip with host rocks. These are medium to coarse-grained at central parts with progressively fine-grained towards the chilled margins. A coarse-grained dolerite sample (TP19) was collected from an N77°E trending dyke near Ontillu village, located near the southern peripheral region of Cuddapah basin for Pb—Pb baddeleyite dating. It should be noted here that paleomagnetic results have been reported from this same location (Site 31) by Belica et al., 2014. At least three cores of 2.5-in. diameter were drilled from each block samples at rock cutting section, CSIR-National Geophysical Research Institute, Hyderabad (CSIR-NGRI). Later, these cores were cut into cylinder specimen of 2.2 cm in length, and at least six cylindrical specimens were obtained from each sample.

4. Methodology

4.1. Pb—Pb Baddeleyite Geochronology

For Pb—Pb Baddeleyite Geochronology, the methodology adopted by Kumar et al., 2014 was followed and is discussed briefly here. About ~500 g of sample (TP19) was crushed to <125-μm grain size with the help of jaw crusher and ring grinder. The powdered sample was concentrated using the "water-based" separation technique on a Wilfley table (Söderlund and Johansson, 2002) at rock crushing section, at CSIR-NGRI. Best quality Baddeleyite grains (mineral inclusions, fractures, and alteration free) were handpicked, after a careful visible inspection under a binocular microscope, from this concentrated fraction. The selected fractions, comprising of 15–30 grains and weighing about 4 and 12 μg (depending on the grain size), were bathed with 4 N HNO_3 for two hours, at room temperature, with sporadic ultra-sonication in a Teflon beaker. Afterwards, the grains were washed several times with MQ water and then left to dry.

0.030 in. wide and 0.001 in. thick, zone refined rhenium filaments were used to load the baddeleyite grains. These filaments were welded to the loading posts. The filaments were bent using a jig, to form a transverse 'U' shaped valley (0.5 mm broad and 0.5 mm deep) in the centre. It was followed by outgassing and cleaning of 'U' shaped

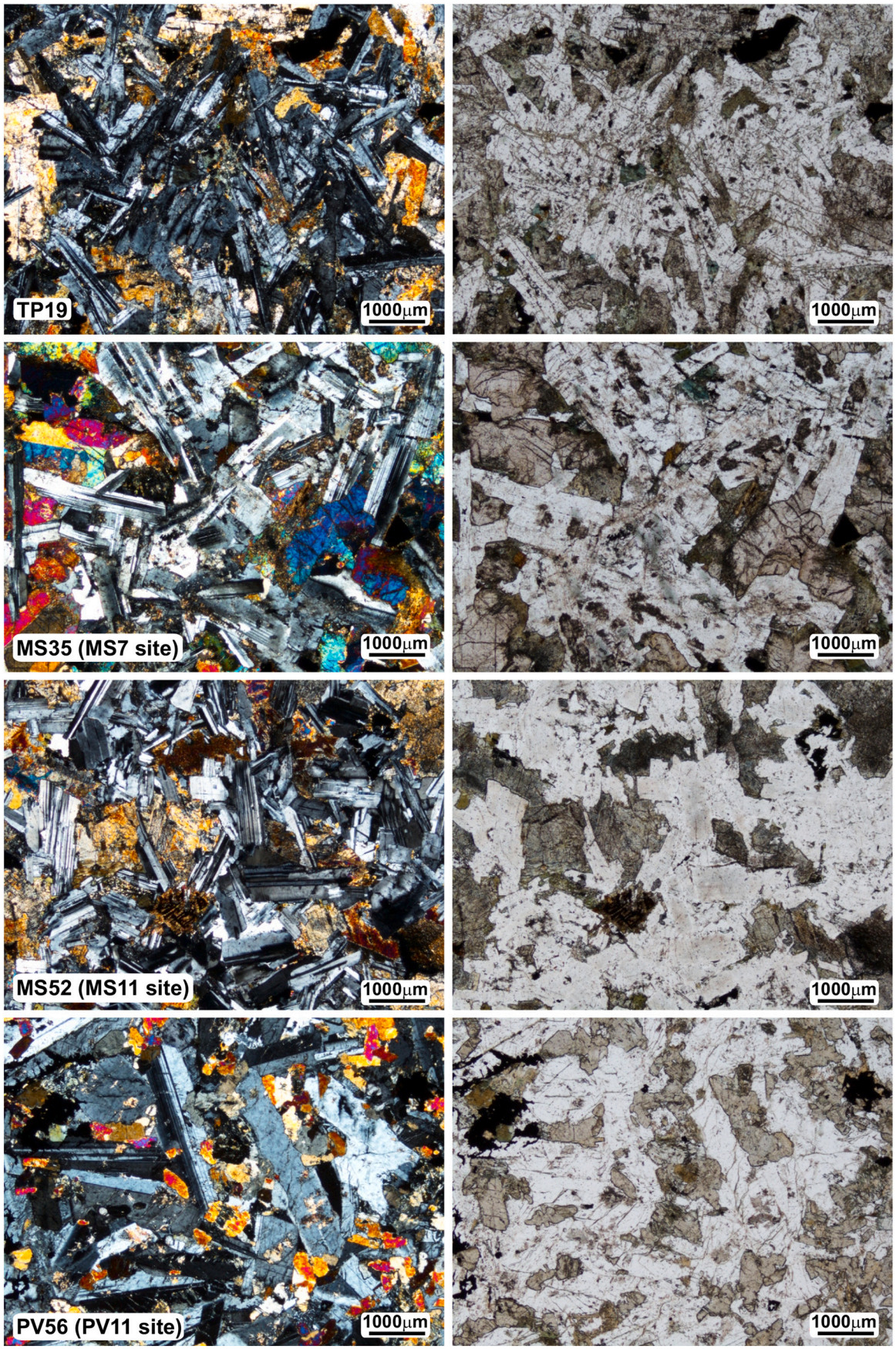


Fig. 2. Photomicrographs are showing mineral assemblage of representative samples (under XPL and PPL) from ~1886 Ma and ~ 1864 Ma dykes and sills.

Table 1
TE-TIMS Pb isotopic data on baddeleyite fractions from the ENE-WNW dyke, Dharwar craton (Ludwig, 2003).

Sample. No.	No. bloc.	$^{206}Pb/^{204}Pb$ (m)	Abs. err	$^{207}Pb/^{206}Pb$ (m)	$^{207}Pb/^{206}Pb$ (c)	Age (Ma)
TP19 (13.380°N; 79.418°E), Strike: N77°E						
TP19–1	27	13,786	319	0.115149 ± 110	0.114163 ± 114	1866.7 ± 2.8
TP19–2	26	13,171	243	0.115160 ± 061	0.114124 ± 065	1866.1 ± 2.1
TP19–3	30	16,942	397	0.114968 ± 056	0.114159 ± 061	1866.6 ± 2.0
TP19–4	25	16,335	1134	0.115172 ± 055	0.114255 ± 036	1868.2 ± 1.6

Weighted Mean age = 1867.1 ± 1.0 Ma, MSWD = 1.02

filaments at alternating high (4.0 A) and low (2.0 A) temperature steps, with the complete sequence of 90 min' duration. The cleaned and outgassed filaments were for at least a week before loading. The grains were loaded carefully onto the bottom of the valley of these filaments. The loaded grains were kept in place by adding a minute droplet of silica gel for annealing at 1250 °C in a vacuum chamber (at ~1*10^{-7} mbar) for 30 min. It helps in expelling disturbed Pb from altered domains. Subsequently, these grains were embedded in silica glass by fusing them with concentrated silicic acid and 9 N phosphoric acid mixture (in 2:1 ratio) at 1200 °C for ~1|0 min, in the vacuum chamber (Fig. 2), on the same Re filament. To keep blank at minimum (<2 pg), less than one µl of silica gel mixture was used. All the loading on filament was done under a stereomicroscope placed in a clean air station equipped with HEPA filter. The TRITON^{PLUS} Thermal Ionization Mass Spectrometer was used for analysis at the CSIR-NGRI. The procedure for the collection of data and data reduction was followed as given in Kumar et al., 2014.

4.2. Paleomagnetic studies

A total of 259 specimens (combining dykes and sills samples) from 26 sites (minimum four specimens per site) were subjected to standard alternating field demagnetisation (AF) using Molspin Alternating Field demagnetiser and thermal (TH) demagnetisation using MMTD 80 oven (Magnetic Measurements limited). The AF demagnetisation included following steps viz., 2.5 mT increments for 5 to 17.5 mT interval, 5 mT increments for 20 to 30 mT interval and 10 mT increments for 40 to100 mT interval. The TH demagnetisation (TH) was performed from 50 to 500 °C with 50 °C increments and from 500 °C to 600 °C in 20 °C increments. The natural remanent magnetisation (NRM) was measured after each incremental demagnetisation step with the help of JR-6 spinner magnetometer (Agico). The Principal Component Analysis (Kirschvink, 1980) has been done for obtained directional data utilising of Zijderveld diagrams (Zijderveld, 1967) follow of standard paleomagnetic protocols.

Moreover, conventional Isothermal Remanence Magnetisation (IRM) acquisition studies were carried out on at least one sample from each site using the Pulse magnetiser. We followed the procedure of incremental magnetic field up to 500 mT and also for the backfield. After each incremental magnetic field step, the NRM results were obtained by JR-6 spinner magnetometer. Rock magnetic and Paleomagnetic measurements were carried out at Paleomagnetism laboratory, CSIR-NGRI.

5. Results

5.1. Petrography

Petrographic studies carried out on at least one sample from each site, and the results are shown as representative photomicrographs in Fig. 2. The ophitic texture is very commonly observed in most of the samples, though the porphyritic texture was also observed in a few samples. Most samples contain plagioclase (mostly labradorite) and augite as principal mineral constituents with minor amounts of (3–5%) of opaque minerals. All the examined samples are fresh, minor alteration of plagioclase and pyroxenes also observed. Magnetic minerals like magnetite and Ti- magnetite occur as opaque in studied samples.

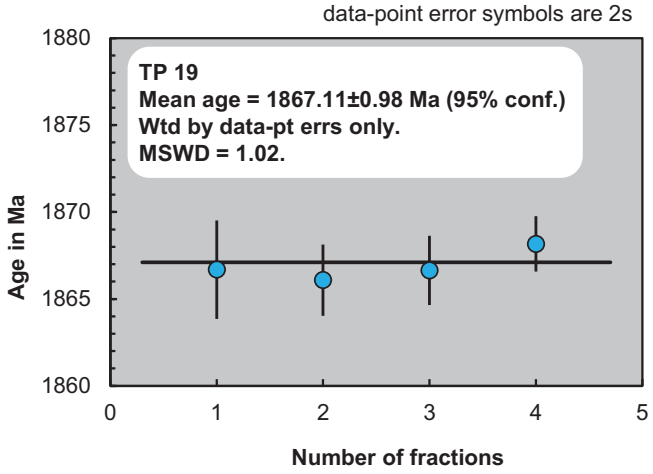


Fig. 3. TE-TIMS weighted mean $^{207}Pb/^{206}Pb$ ages on four baddeleyite fractions from TP-19 sample (location given in Fig. 1). Error bars represent 95% confidence limits of measurements.

5.2. Pb—Pb baddeleyite geochronology

The coarse-grained dolerite sample (TP 19) was collected from N77°E trending dyke near Ontillu village, located in the densest concentration of dykes in the southern region to the Cuddapah basin (Fig. 2). This sample yielded ample prismatic, inclusion and alteration free baddeleyite grains (80–110 µm). The Pb isotopic results of the analysis on four baddeleyite fractions are given as Table 1 and as illustrated in Fig. 3. Each fraction contained minimum 25 blocks. All four fractions were showing similar concentrations of Pb isotope ratios. The weighted mean of ^{207}Pb—^{206}Pb ratios yielded an age of 1867.1 ± 1.0 Ma (MSWD = 1.02).

Recently, three similar Pb—Pb baddeleyite ages were reported from the Eastern Dharwar Craton, which included two mafic sill samples (PV63 and PV73) intruding into Tadipatri formation of the Cuddapah basin (1863.6 ± 0.9 Ma, 1864.3 ± 0.6 Ma; Sarma, 2020) and one dyke sample (MSG10) from the northern periphery of Cuddapah basin (1861.8 ± 1.4 Ma, Sarma, 2020). This mafic episode is the second to the youngest mafic magmatic event from the Dharwar craton and known as SangalaPalli-Chimiryala swarm (Sarma, 2020). The age determinations of Sarma (2020) along with age reported here, yielded a weighted mean age of 1864.4 ± 2.7 Ma. This can be considered as the best estimation of emplacement age for Sangala Palli-Chimiryala swarm.

5.3. Isothermal Remanent Magnetization (IRM)

Results of IRM study for these samples are shown as typical saturation curves (Fig. 4-a and b). The saturation of magnetisation in these rocks was attained between 150 mT and 200 mT. The backfield coercivity of remanence values ranges from 40 mT to 50 mT. The results of IRM studies indicate that the magnetic remanence for these samples has been carried by Pseudo single domain or Single domain (i.e., PSD or SD)

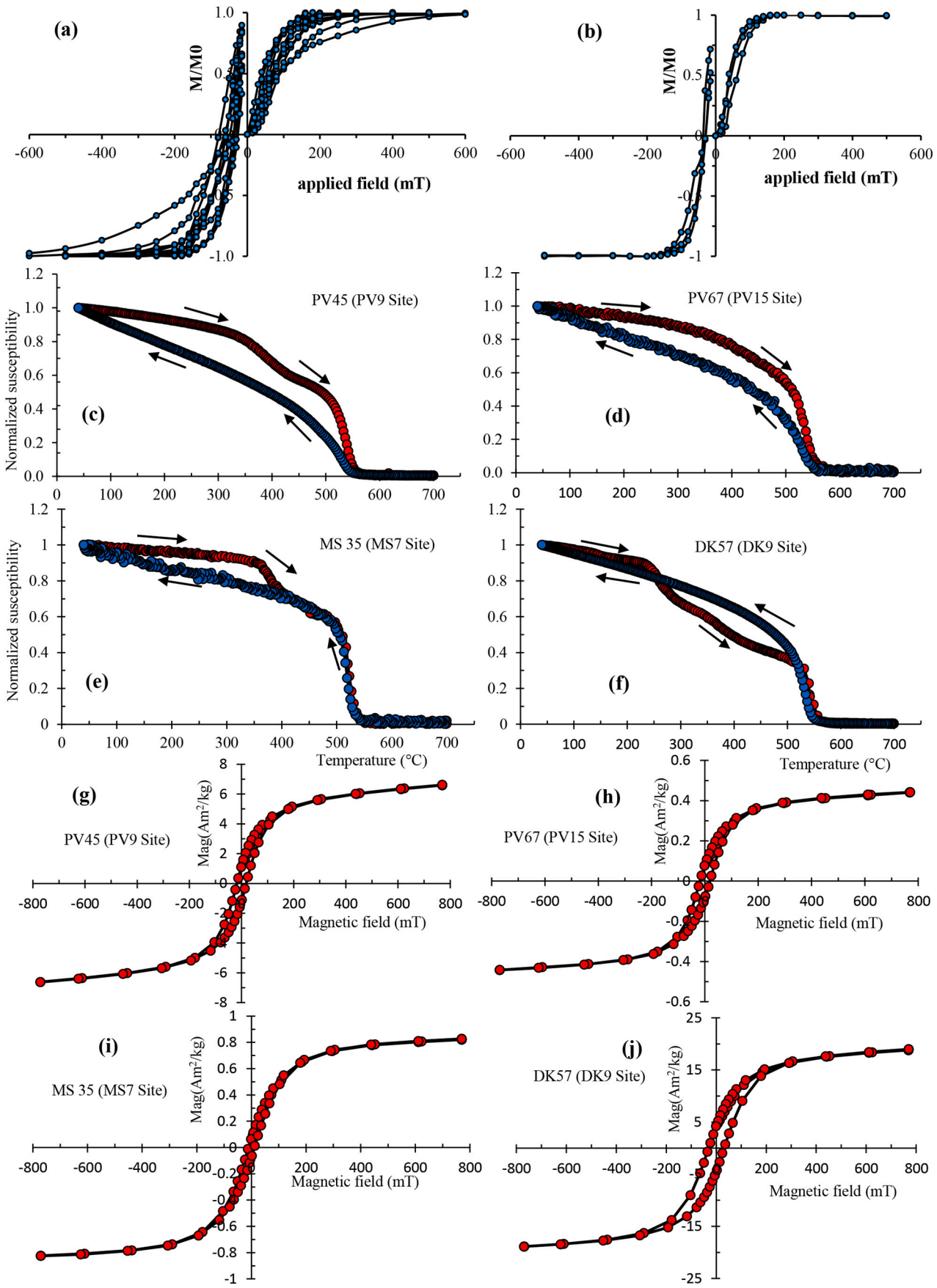


Fig. 4. Typical examples of Isothermal remanence acquisition curves and back field IRM for ~1886 Ma samples (a) and ~ 1864 Ma samples (b). Magnetic susceptibility versus temperature dependencies curves (c to f) and Hysteresis loops (g to j) for representative samples from each swarm.

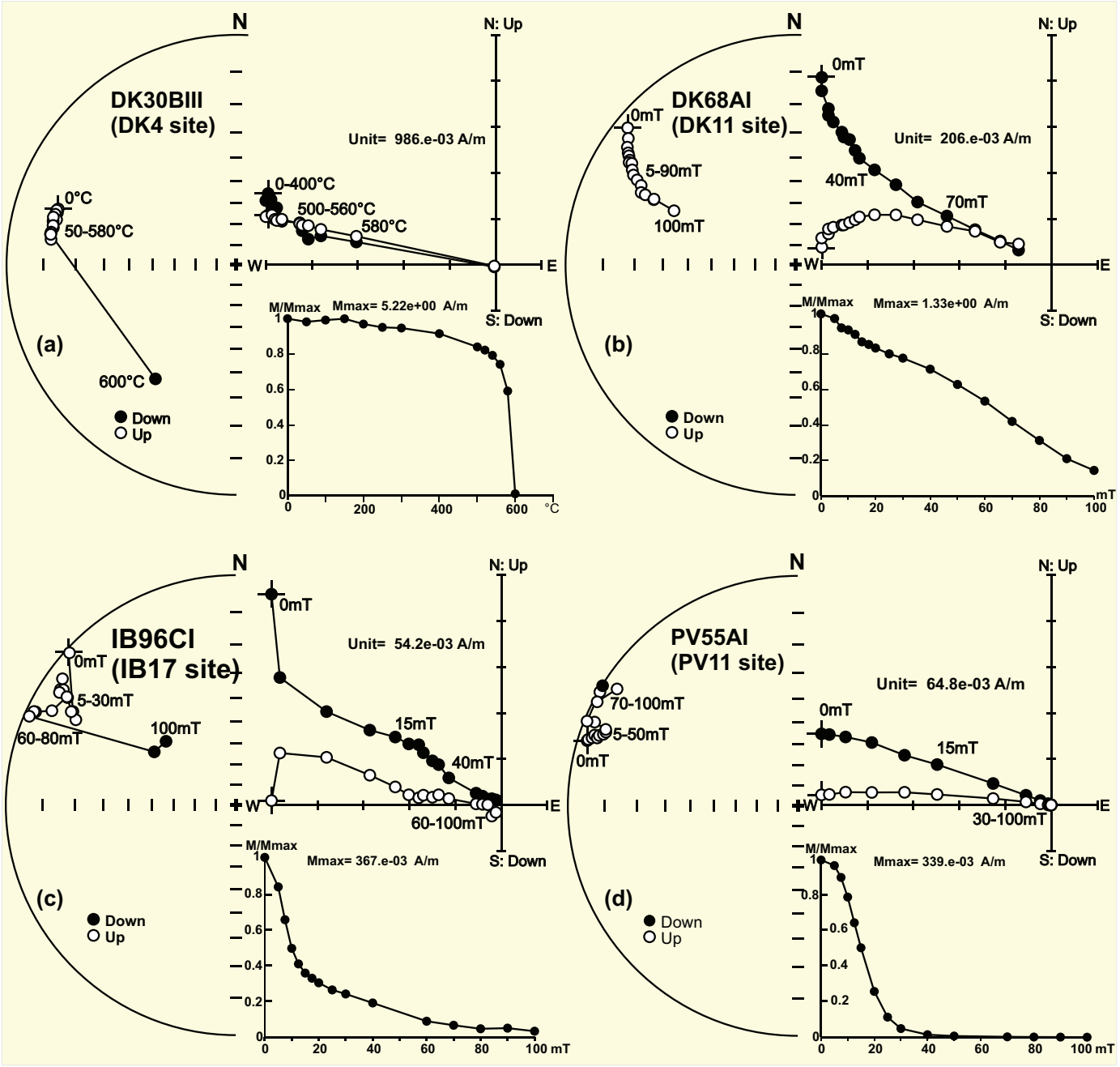


Fig. 5. Zijderveld diagrams and equal area stereonet projections showing characteristic behavior of natural remanence to thermal demagnetization (a) and AF demagnetization (b to d) for representative samples from ~1886 Ma sites. Thermal demagnetization measurements are in °C and AF measurements are in millitesla (mT). Open/closed circles in the stereoplots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0, a plotting and analysis program (Chadima and Hrouda, 2006).

magnetite grains. The thermomagnetic curves of low field susceptibility for representative samples shows curie temperature is close to 580 °C and suggest that the magnetic remanence carrier of these samples is Ti-poor magnetite (Fig. 4c to f). Hysteresis loops studies is also display the presence of pseudo-single domain titanomagnetite or magnetite in the studied samples (Fig. 4g to j).

5.4. Paleomagnetism

The paleomagnetic results of this study have revealed two distinct groups of characteristic remanent magnetic (ChRM) directions (Figs. 5, 6, and 7; Table 2). Six of all sites (published as well as this study) reported here (Table 2) have precise radiogenic ages reported on them along with the ChRM directions. Correlation of ChRM data from such sites with all others, identify them to distinct mafic magmatic episodes at ~1886 Ma and ~ 1864 Ma (Fig. 5). Hence the results of the paleomagnetic analysis for these events are discussed below.

5.4.1. 1886 Ma swarm

The paleomagnetic results on a total of 169 specimens representing 22 mafic dykes (140 specimens) and mafic sills (4 sites, and 29 specimens) of ~1886 Ma swarm are presented here. Site locations and results are given in Table 2, and typical demagnetisation projections are displayed in Fig. 5. In general, the intensity of NRM for most of the samples

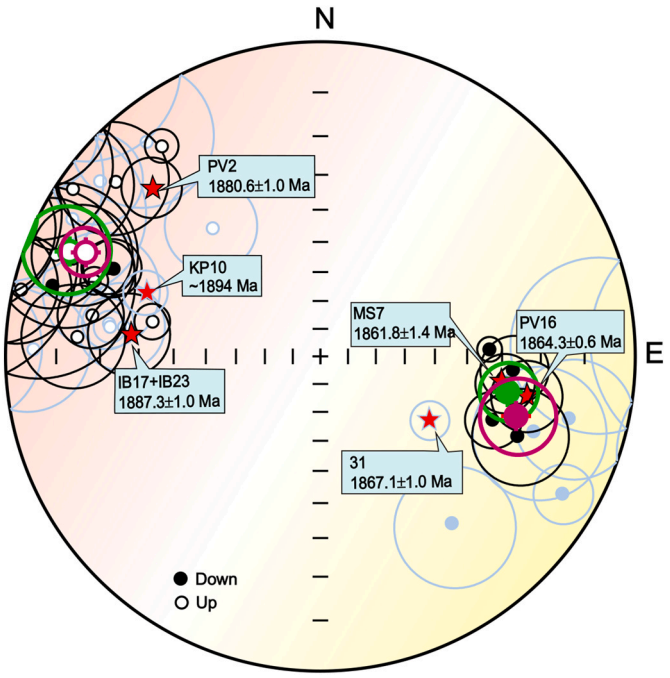


Fig. 6. Stereo plot showing paleomagnetic data, which shows site mean characteristic remanence directions with ovals of 95% confidence from the ~1886 Ma and ~ 1864 samples. Present study site mean directions are represented as black circles and reported site mean directions are shown by grey circles (Table 2). Red stars with black or grey 95% confidence circles show Precise ages with paleomagnetic directions of Present study or published results respectively. Pink filled/ open circles represent mean direction of Present study and grand mean of all sites (Present and published sites) represents green filled/ open circles for the respective events.

is strong and uniform. We used both alternating field (AF) and stepwise thermal demagnetisation (TH) techniques to isolate the characteristic remanent magnetisation directions. All the demagnetised samples showed fairly consistent decay behaviours to origin component and yielded consistent magnetic directions. The samples show stable ChRM with high unblocking temperatures (400–560 °C) during TH demagnetisation and high coercivity component (40–90 mT) in the AF demagnetisation. Samples from 20 sites yielded uniform ChRM directions with north-westerly declinations and shallow inclinations (Fig. 6, Table 2). The paleomagnetic data from three locations (HY23, MS16, DG13) were excluded from mean direction calculations, as they were probably located on older dykes (Radhakrishna et al., 2013b; Kumar et al., 2015; Nagaraju et al., 2018b) and hence were suspected of carrying the overprinted directions. Samples at HY 19 site yielded normal and reverse magnetic polarity directions. The directions of sills were used in mean calculation after applying the bedding dip correction of 15° to the measured data (Clark, 1982). The mean direction calculated for 17 sites (13 dykes and 4 sill sites) has a declination = 293° and inclination = −9° (k = 17, α_{95} = 9°) with a resultant paleopole at 21°N and 338°E (dp = 5°, dm = 9°).

The magnetic directions from this study reasonably match with the published results on mafic dykes from different regions of Dharwar craton. Such results include data from Tirupati dykes (Anjanappa, 1972), Cuddapah dykes (Prasad, 1966), Kolar dykes (Reddy and Prasad, 1979), mafic traps from Cuddapah basin (Clark, 1982), Tiptur dykes (Bhalla et al., 1980), Anantapur dyke (Kumar and Bhalla, 1983), mafic dykes around Tirupati region (Goutham et al., 2011), mafic dykes around Cuddapah basin (Radhakrishna et al., 2013b), 1.88 dykes (Belica et al., 2014), and Kamalapur dyke at KP 10 site (Nagaraju et al., 2018a). Locations of all these studies are also shown in Fig. 9, and statistical results are summarised in Table 2. Majority of these sites were distributed around the Cuddapah basin in the EDC and Tiptur region of WDC with dominant orientations of *E*-W and NW-SE dykes. All these directions were overlapping within errors and along with data of present study yielded a normal polarity direction with a grand mean of declination = 294°, inclination = −12°, (K = 11, α_{95} = 5°), and corresponding paleopole at 21°N and 336°E (dp = 3°, dm = 5°) (Fig. 8).

5.4.2. 1864 Ma swarm

Altogether 65 specimens from seven sites, including three mafic dykes (35 specimens) and sills (at 3 sites, 30 specimens) of ~1864 Ma swarm are presented here. Most of these samples show consistent NRM intensities for all the dykes and sills. High blocking temperature (TH) and high coercivity (AF) nature of component reveal stable magnetic directions (Fig. 7) with a south-easterly declination and shallow inclination (Fig. 6). These directions exhibit reversed polarity magnetisation relative to ~1886 Ma directions. However, within site mean directions of all seven sites of this event yielded a mean of declination = 101°, inclination = 28° (k = 76, α_{95} = 8°) with a corresponding paleopole at 6°S, 152°E (dp = 5°, dm = 9°).

Similar magnetic remanence directions have been reported earlier from several mafic intrusions adjacent to Cuddapah basin (Radhakrishna et al., 2013b; Belica et al., 2014). Such results are given in Table 2, and locations are shown in Fig. 9. Combined paleomagnetic direction from the present study together with the previously reported data gave a mean declination = 107°, inclination = 24°, (k = 21, α_{95} = 10°). The corresponding paleopole lie at 13° S and 152° E (dp = 6°, dm = 10°) (Fig. 8).

5.4.3. Reliability of primary magnetisation

Although, we were not able to collect baked contact test samples during this fieldwork to prove the reliability of primary magnetisation in these rocks, following evidence supports that the reported remanence directions of this study are primary and were acquired during the time of their emplacement.

i) The ~1886 Ma and ~ 1864 Ma dykes and sills yielded precise U-Pb/Pb-Pb ages (constrained to less than ±5 Ma) coupled with stable paleomagnetic directions (with α_{95} = 5° and 10°, respectively). The ~1886 Ma paleomagnetic results are overlapping with earlier reported normal polarity magnetization results, which were supported by positive bake contact test in this region (Belica et al., 2014).

ii) The stepwise demagnetization effectiveness is confirmed by multiple demagnetization methods (AF, and TH demagnetisation). We have carried out the paleosecular variation test (PSV) for both ~1886 Ma and ~ 1864 Ma poles. The calculated VGP scatter value (Cox, 1970) for the seventy-nine sampled locations from ~1886 Ma swarm is 17.5° + 0.8°/−9.8° and twelve sites of ~1864 Ma swarm is 15.6° + 2.5°/−7.7°, distinctly suggests that PSV has been averaged out for these respective poles.

iii) Further, remanence carrier of magnetization has been determined as pseudo-single titanomagnetite or magnetite in these rocks.

iv) The preservance of primary magnetization supported by positive baked contact test and dual-polarity magnetization by the older mafic intrusions at ~2367 Ma, ~2216 Ma, ~2207 Ma from this region, suggest that the rocks of this sector have not been thermally heated above their blocking temperatures. And hence preserve the primary nature of remnant magnetization.

v) The obtained paleomagnetic poles of 1886 Ma and 1864 Ma swarms are satisfying the Meert et al. (2020) reliability criteria value *R* = 6 and 5, respectively. Hence, they can be considered as primary and key paleomagnetic poles.

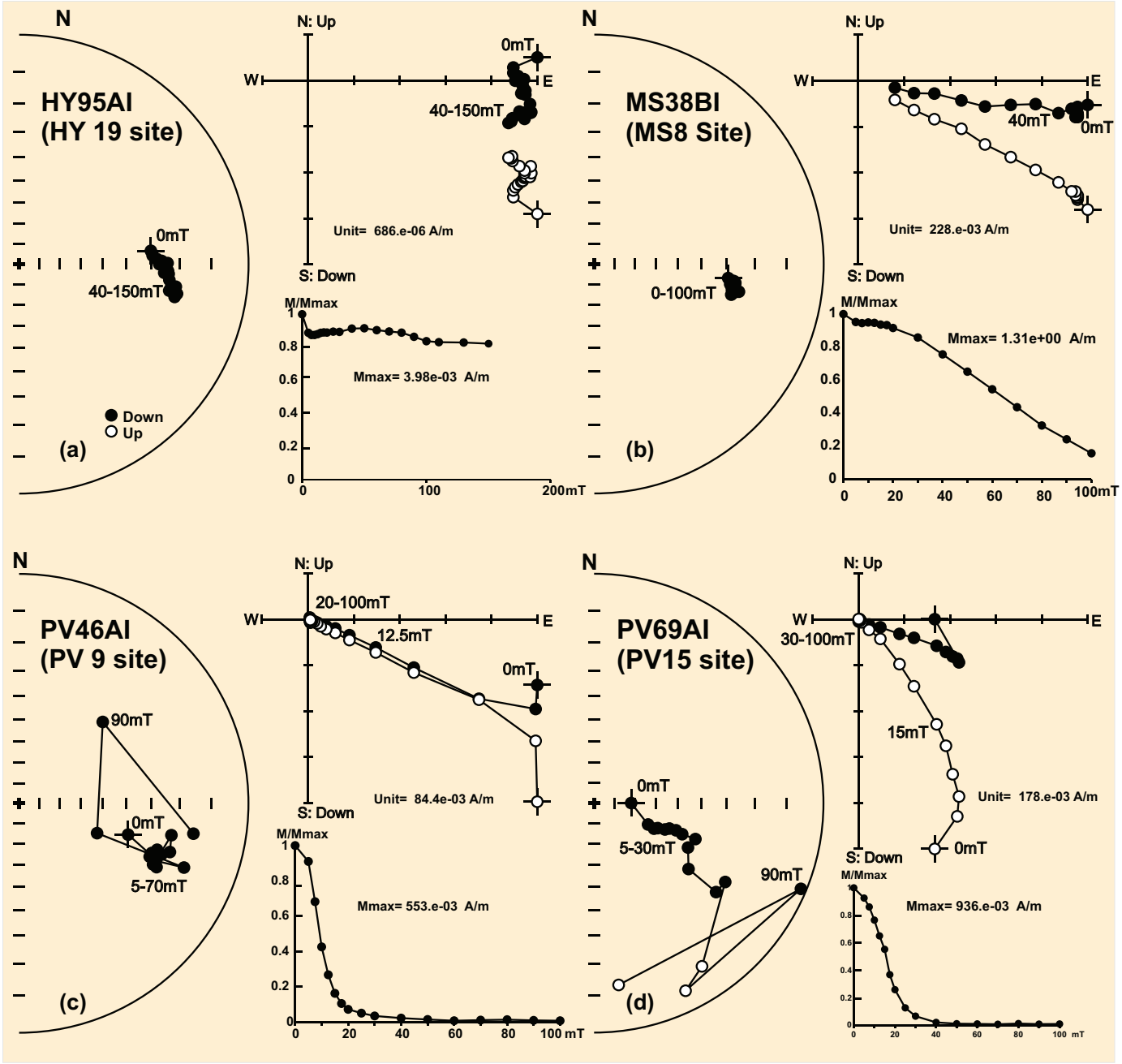


Fig. 7. Zijderveld diagrams and equal area stereonet projections showing characteristic behavior of natural remanence to AF demagnetization for representative samples from ~1864 Ma sites. AF measurements are in millitesla (mT). Open/closed circles in the Steroplots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6. Discussion

6.1. New key paleomagnetic pole at ~1864 Ma

The predominant geographical distribution of almost *E*-W trending dykes below the Cuddapah basin was inferred as a part of ~2.37 Ga and ~ 1.89 Ga dyke swarms by earlier studies based on their favourable orientation (Halls et al., 2007; French and Heaman, 2010; Kumar et al., 2012a) and paleomagnetic directions (Radhakrishna et al., 2013b; Belica et al., 2014). However, the present study reports a precise Pb—Pb baddeleyite age of 1867.1 ± 1.0 Ma for an N77°E striking dyke from this region and hence identified a distinctly younger magmatic event, directly discarding the popular understanding in this regard. This rigorous age confirmed the existence of ~1864 Ma mafic magmatism in the Dharwar craton. Moreover, along with published synchronous emplacement ages (1861.8 ± 1.4 Ma; 1863.6 ± 0.9 Ma, 1864.3 ± 0.6 Ma; Sarma, 2020), this magmatic episode in the EDC seems to have a large spatial extent of about ~400 km (based on the distance between farthest located MSG10 and TP19 sites of ~1864 Ma; Fig. 9) within the brief time span of ~5 Ma (1862 Ma–1867 Ma) for its emplacement. Further, two dated felsic tuffs (weighted mean age of 1862 ± 9 Ma; Sheppard et al., 2017) from Tadipatri formation of Cuddapah basin supports the presence of ~1864 Ma magmatism in the Dharwar craton (Fig. 9). The emplacement of this event is useful to constrain the upper limit of age estimation for sedimentation in the Tadpatri formation, Cuddapah basin (stratigraphic column given in Fig. 10).

Table 2
Paleomagnetic results on ~1864 Ma and ~ 1886 Ma dykes/sills from the Dharwar craton.

Site	Lat (°N)	Long (°E)	Strike (°E)	n/N	D (°)	I (°)	K	α95	Plat (°N)	Plong (°E)	dp	dm	Ref.
~1864 Ma dykes/Sills													
HY 19	17.343	78.805	86	9	110	30	46	8	-14	147	5	9	1
MS 7*	16.966	80.047	174	10	98	29	34	8	-3	153	5	9	1
MS 8	16.968	80.021	8	8	91	30	211	4	3	154	2	4	1
MS 9	16.961	80.019	22	8	84	37	167	4	12	152	3	5	1
PV 9^{S}	14.714	78.294	–	13	112	22	13	12	−23	146	7	13	**1**
PV 15^{S}	14.736	78.157	–	8	94	27	41	9	−1	153	5	10	**1**
PV 16*^{S}	14.773	78.173	–	9	101	23	34	9	−7	154	5	10	**1**
Mean	**15.916**	**78.916**	**–**	**7/65**	**101**	**28**	**76**	**8**	**−6**	**152**	**5**	**9**	
18	13.068	77.009	330	11	120	7	56	6	28	336	3	6	3
31*	13.379	79.410	255	7	120	46	97	6	19	314	5	8	3
32	13.417	79.413	310	4	142	22	GC	GC	45	318	–	–	3
P5	16.79	77.96	128	8	104	12	19	13	−12	158	7	13	4
P7	16.75	78.13	114	6	109	19	19	16	−15	153	9	17	4
P74	17.14	79.11	10	6	88	10	15	18	4	165	9	18	4
Grand mean	**15.503**	**78.711**	**–**	**13/107**	**107**	**24**	**21**	**10**	**−13**	**152**	**6**	**10**	
~1886 Ma dykes/Sills													
HY 10	17.436	78.720	112	10	283	−2	23	10	12	344	5	10	1
HY 19	17.343	78.805	73	6	311	−9	33	12	37	329	6	12	1
HY 21	17.345	78.816	88	5	285	7	17	19	15	348	10	19	1
HY 23#	17.323	78.846	13	6	298	2	34	12	27	341	6	12	1
IB 17*	17.206	78.618	115	9	276	−30	17	13	1	331	8	14	1
IB 23	17.208	78.602	112	11	277	−25	121	4	2	334	2	5	1
DK 4	16.654	78.915	50	11	289	−16	200	3	15	336	2	3	1
DK 9	16.540	78.818	140	11	282	−33	83	5	6	328	3	6	1
DK 10	16.537	78.808	335	7	293	18	93	6	24	352	3	7	1
DK 11	16.663	79.287	98	9	304	−3	18	13	32	337	6	13	1
DK 28	16.605	79.120	117	9	275	−15	20	12	2	341	6	11	1
MS 11	17.041	79.954	104	8	292	−10	17	15	19	338	8	15	1
MS 16#	16.918	79.430	102	6	271	11	82	7	2	355	4	8	1
MS 21	17.570	79.541	125	5	280	−17	97	8	7	338	4	8	1
DG 1	15.440	77.835	133	10	323	−10	181	4	48	321	2	4	1
DG 13#	15.348	77.816	139	7	287	−5	93	7	16	341	4	7	1
PV 2*^{S}	14.419	78.229	50	7	315	−16	126	7	40	324	4	7	1
PV 3^{S}	14.463	78.137	130	6	293	−12	38	11	21	336	6	11	1
PV 10^{S}	14.607	78.041	130	8	293	19	71	7	24	352	4	7	1
PV 11^{S}	14.457	78.442	50	8	291	−6	28	11	20	340	6	11	1
Mean	**16.270**	**78.755**	**–**	**17/140**	**293**	**−9**	**17**	**9**	**21**	**338**	**5**	**9**	1
KP 10*	15.301	76.481	–	1/10	290	−29	61	6	14	326	4	7	2
1.9 mean	14.499	78.184	–	23/142	307	−7	12	9	34	333	5	9	3
1.99–1.89	15.757	79.128	–	10/61	271	−5	16	12	1	346	6	12	4
ID dykes	13.468	79.342	–	4/28	279	−22	205	6	6	336	4	7	5
Dyke V	14.400	77.700	–	2/28	320	−34	68	13	38	292	8	15	6
Tiptur dykes	13.300	76.500	–	6/35	287	−19	24	14	14	333	8	15	7
Cuddapah traps	14.300	78.200	–	6/48	299	−6	18	16	27	337	8	16	8
Kolar dykes	12.959	78.265	–	2/33	321	−8	296	15	48	327	7	15	9
Tirupati dykes	14.000	79.000	–	3/18	278	−22	221	4	5	336	2	4	10
Cuddapah dykes	13.600	79.360	–	5/30	279	−26	47	11	5	334	7	12	11
Grand mean	**14.821**	**78.446**		**79/573**	**294**	**−12**	**11**	**5**	**21**	**336**	**3**	**5**	

Previously, when this younger mafic event (~1864 Ma) was not recognised in Dharwar craton, the dual polarity magnetic directions were interpreted as a record of ~1886 Ma magmatism (Radhakrishna et al., 2013b; Belica et al., 2014). Reliable paleomagnetic directions from this study reveal that the reverse polarity remanent magnetic directions were resulted by the emplacement of newly identified ~1864 Ma mafic event. The mean paleomagnetic results from 12 sites yield Declination =107°; Inclination =24° (k = 21; α_{95} = 10°), and corresponding pole situated at 13°S and 152°E (dp = 6°, dm = 10°). This pole is derived from a precisely dated dykes/sills, and satisfies the Meert et al. (2020) reliability criteria value $R = 5$ (1110101). Hence the ~1864 Ma pole reported here can be considered as a key paleomagnetic pole, and provides an opportunity to ascertain the paleogeographic position of the Dharwar craton within supercontinent configuration during Paleoproterozoic.

The coeval ~1864 Ma mafic units have not been reported from any other Indian cratons. Recently, a slightly younger age (1851.1 ± 2.6 Ma, Shellnutt et al., 2019) was reported for NNW striking Sonakhan mafic dyke from the northern Baster craton. Preliminary paleomagnetic results reported from different parts of the Bastar craton (Meert et al., 2011; Radhakrishna et al., 2013a), show both normal and reverse magnetic directions for most of the NNW-SSE trending dykes, which were considered as part of ~1886 Ma LIP (Meert et al., 2010; Radhakrishna et al., 2013a). However, the calculated reverse magnetic polarity mean direction of the Bastar craton (data are taken from Meert et al., 2010; Radhakrishna et al., 2013a) is reasonably matching (within error) with the ~1864 Ma paleopole from the Dharwar craton (Fig. 8). It hence is suggestive towards the presence of still unknown reverse polarity magnetisation probably acquired by ~1864 Ma event in the Bastar craton.

6.2. Update the key paleomagnetic pole at ~1886 Ma

Four precise U-Pb/Pb-Pb baddeleyite age determinations have established the emplacement age of ~1886 Ma swarm in the Dharwar craton. These include age determinations from Pulivendla sill of Cuddapah basin (1880.6 ± 1.0 Ma and 1885 ± 3.1 Ma; French et al., 2008 and Sarma, 2020 respectively), N100°E dyke near Ibrahimpatnam

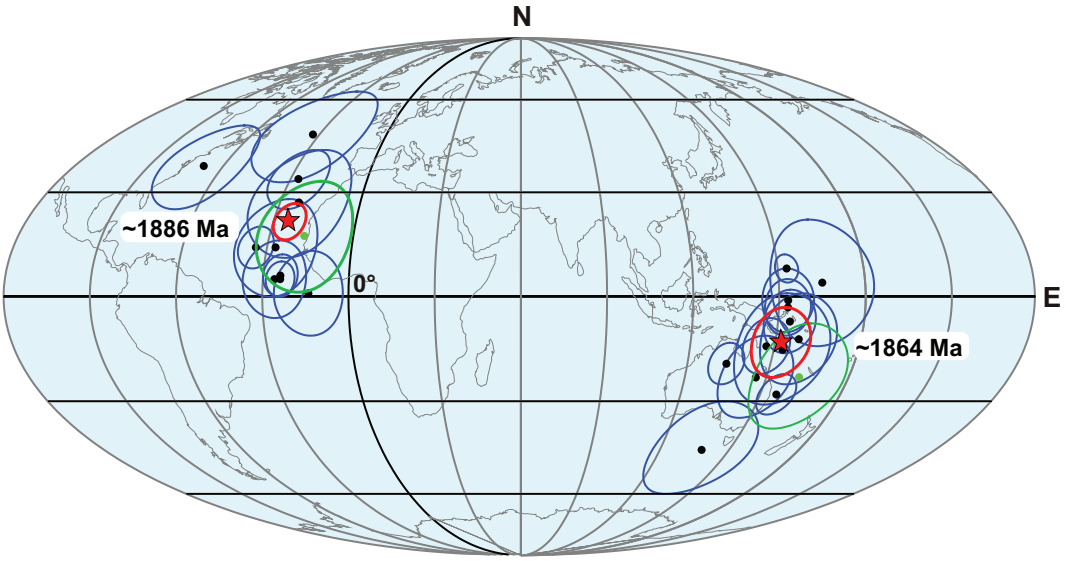


Fig. 8. Mollweide projection showing the virtual geomagnetic poles of site means from the ~1886 Ma and ~ 1864 Ma events. Blue circles of ~1886 Ma/1864 Ma represent paleomagnetic pole position of present and reported sites grand mean/ site means respectively (Table 2). Grand mean pole is shown with a red filled star for respective events. Shown in open green circles are paleopoles of the sites from Bastar craton, calculated from Meert et al. (2011) and Radhakrishna et al. (2013a) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(1887.3 ± 1.0 Ma; Sarma, 2020), and Hampi dyke (~1894 Ma, Halls et al., 2007). The calculated weighted mean of all these intrusions yield an age of 1886 ± 5 Ma and also suggests a rapid emplacement within ~13 Ma (1881–1894). We have obtained normal polarity magnetisation directions with $\alpha_{95} \leq 8°$, from these dated dykes (at sites IB17, IB23, PV2 (this study) and KP10 (Nagaraju et al., 2018a)). The primary remanence of these dykes has been well demonstrated with positive bake contact test (Belica et al., 2014). We have calculated the mean ChRM directions for ~1886 Ma mafic magmatism using results of present and reported studies from 79 sites. The mean ChRM directions yield D = 294°, I = -12°, k = 11 and $\alpha_{95} = 5°$, with the corresponding paleopole placed at 21°N and 336°E (dp = 3°, dm = 5°). This updated palaeopole is fulfilling the Meert et al. (2020) reliability criteria R value = 6 (1111101) and can be regarded as key palaeopole to be used for determining paleoposition of Dharwar craton at ~1886 Ma.

The contemporaneous mafic magmatism at ~1886 Ma also been found in Bastar craton linked with Bastar- Cuddapah LIP. It comprises of mafic dykes (BD2 swarm) from southern Bastar craton (1891.1 ± 0.9 Ma, 1883.0 ± 1.4 Ma; French et al., 2008), and Bhanupratappur dyke from central Bastar craton (1882.4 ± 1.5 Ma, Shellnutt et al., 2018). A radial emplacement of dykes (NW-SE dykes at Bhanupratappur region to ENE dykes below the Cuddapah basin) is suggested with foci of mantle plume centre lying in the region of Eastern part to EDC by several recent studies (Ernst and Srivastava, 2008; French et al., 2008; Belica et al., 2014; Shellnutt et al., 2018; Stark et al., 2017; Liu et al., 2018). So, considering the similarity in radial strike patterns of ~1864 Ma dyke swarm (NNW-SSE Sonakhan dyke to ENE to WSW Ontillu dyke), we postulate that the same plume source could be responsible for the generation of later ~1864 Ma swarm as an individual pulse in the time period not more than 35 Ma. Additional ages, paleomagnetic and Anisotropy of Magnetic susceptibility studies from this swarm are required to affirm this inference.

Recently, Nagaraju et al. (2018b) have demonstrated an Apparent polar wander path (APWP) and drift rates of Dharwar craton using well defined key paleopoles from 2367 Ma to 1886 Ma period. Here, we have updated the APWP of DC adding a new key paleopole at ~1864 Ma with a drift rate of 5.42 cm/yr during 1886–1864 Ma. It is fair to suggest a normal relative movement from 6°N to 13°S paleolatitude during ~20 Ma (1885–1865 Ma) time interval.

6.3. Global age correlations

The coeval ~1886–1864 Ma mafic magmatism has been identified in several cratons over the worldwide, includes Boonadagin dykes (1888 ± 9 Ma; Stark et al., 2019) and Yalgoo dykes (1852 ± 12 Ma; Ameen and Wilde, 2006) from Australia craton; Molson dykes and related magmatic events (1877 + 7/−4 Ma; 1883 ± 2 Ma; 1883.7 + 1.7/−1.5 Ma;1882.9 + 1.5/−1.4 Ma; Halls and Heaman, 2000; Heaman et al., 1986) in Superior craton; Ghost dykes (1887 ± 5 Ma; Buchan et al., 2016) and Mara river sills (1870 ± 1 Ma; Davis et al., 2004) of Slave craton; Mashonaland sills (1877.4 ± 2.2 Ma; 1885.9 ± 2.4 Ma; 1875.6 ± 1.6 Ma; 1871.9 ± 2.2 Ma; 1882.7 + 1.6/−1.5 Ma; Soderlund et al., 2010; Hanson et al., 2011) in the Zimbabwe craton; Waterburg sills (1878.8 ± 0.5 Ma; 1871.9 ± 1.2 Ma; Hanson et al., 2004), Black hills dyke swarm (1870.7 ± 4.4 Ma, 1862.3 ± 6.5 Ma, 1860.9 ± 2.7 Ma, 1857.0 ± 8.9 Ma; Olsson et al., 2016)of Kaapval craton; Gabbroid bodies of Vittangi (1886 ± 14 Ma), Kiuruvesi (1886 ± 5 Ma), Pohjanmaa-Ylivieska (1879 ± 5 Ma), Jalokoski (1871 ± 4 Ma; Pesonen et al., 2003 and cited in Antonio et al., 2017) and Keuruu dykes (1867 ± 8 Ma; 1870 ± 9 Ma; Klein et al., 2016) in Baltica shield; Rhyolitic dykes and flows (1877.4 ± 4.3 Ma; 1853.7 ± 6.2 Ma; 1895 ± 11 Ma; Antonio et al., 2017), Rio maria dykes (1884.6 ± 1.6 Ma; 1880.2 ± 1.5 Ma; Teixeira and Hamilton, 2019) from Amazonia craton; Yinjiaping mafic dyke (1866 ± 21 Ma; Qiu et al., 2020) in the South China craton; Eastern Murman sill province (1860 ± 4 Ma; Veselovskiya et al., 2019) in Murmansk craton of Kola Karelian craton, Kalaro-Nimnyrsky dyke swarm (1869 ± 2 Ma; Ernst et al., 2016) and Malazadoisky dyke (1863 ± 1 Ma; Ernst et al., 2016) from Aldan shield of Siberia craton. Apart from these mafic intrusions, dacite (1878 ± 4 Ma) and rhyodacite (1863 ± 9 Ma; Didenko et al., 2009) from Akitkan group of Siberia craton and felsic tuff beds (1891 ± 8 Ma; 1890 ± 10 Ma and 1885 ± 18 Ma; Rasmussen et al., 2012) in Frere iron formation of Western Australia craton also been reported. The widespread of this mafic magmatic episode (~1886–1864 Ma) indicates great crustal extension and simultaneous mafic magmatism across the globe at that time.

6.4. Paleogeographic implications

Paleopositon of India is constrained around the equator at about ~1885 Ma in several paleogeographic reconstruction models proposed

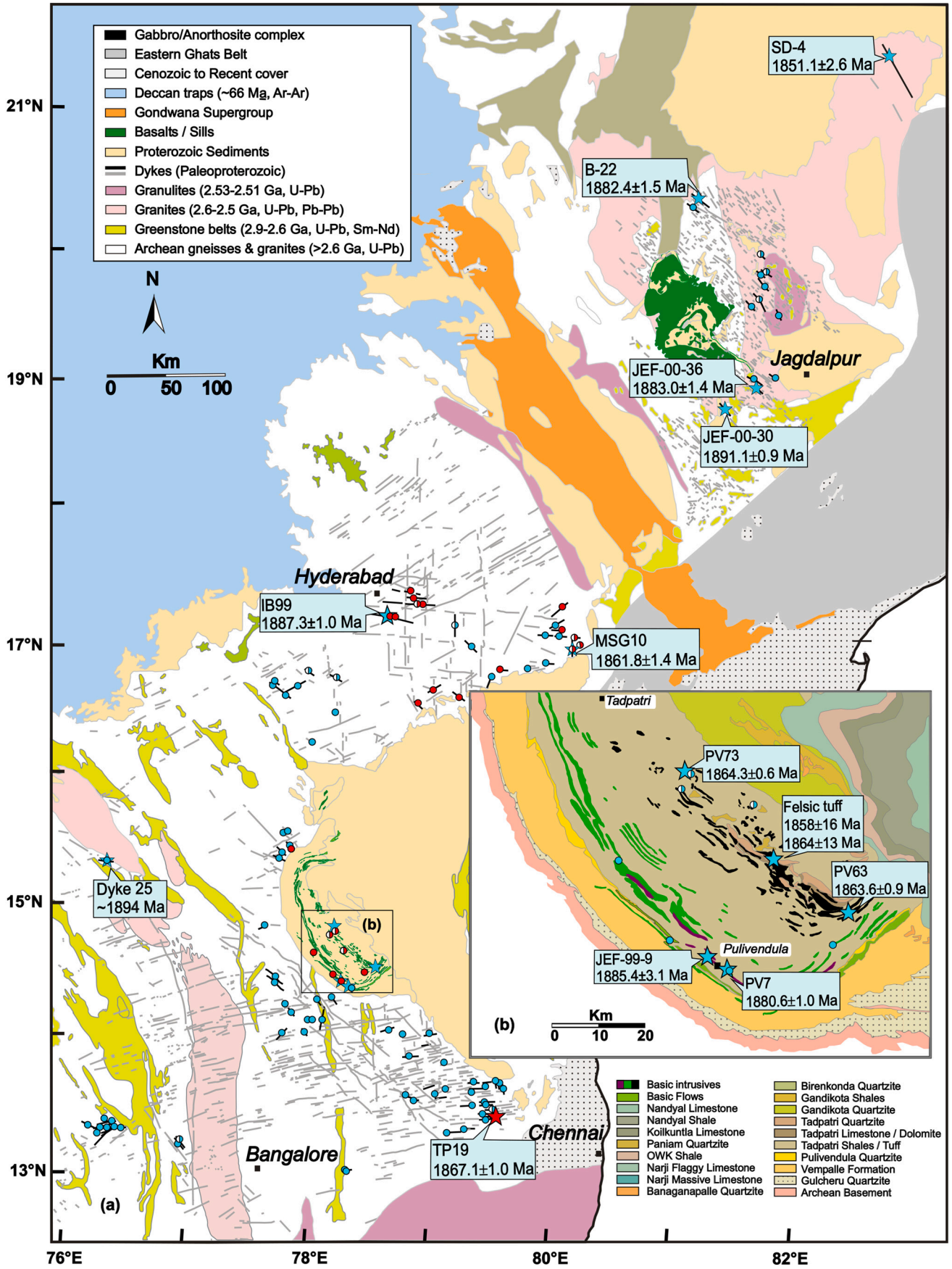


Fig. 9. Modified geological map of French et al., (2008), showing the distribution of mafic dykes along with sampled locations in the Dharwar and Bastar cratons. Red/cyan blue filled circles shows ~1886 Ma sites and semi filled circles of red/ cyan blue indicates present and published sites of ~1864 Ma.

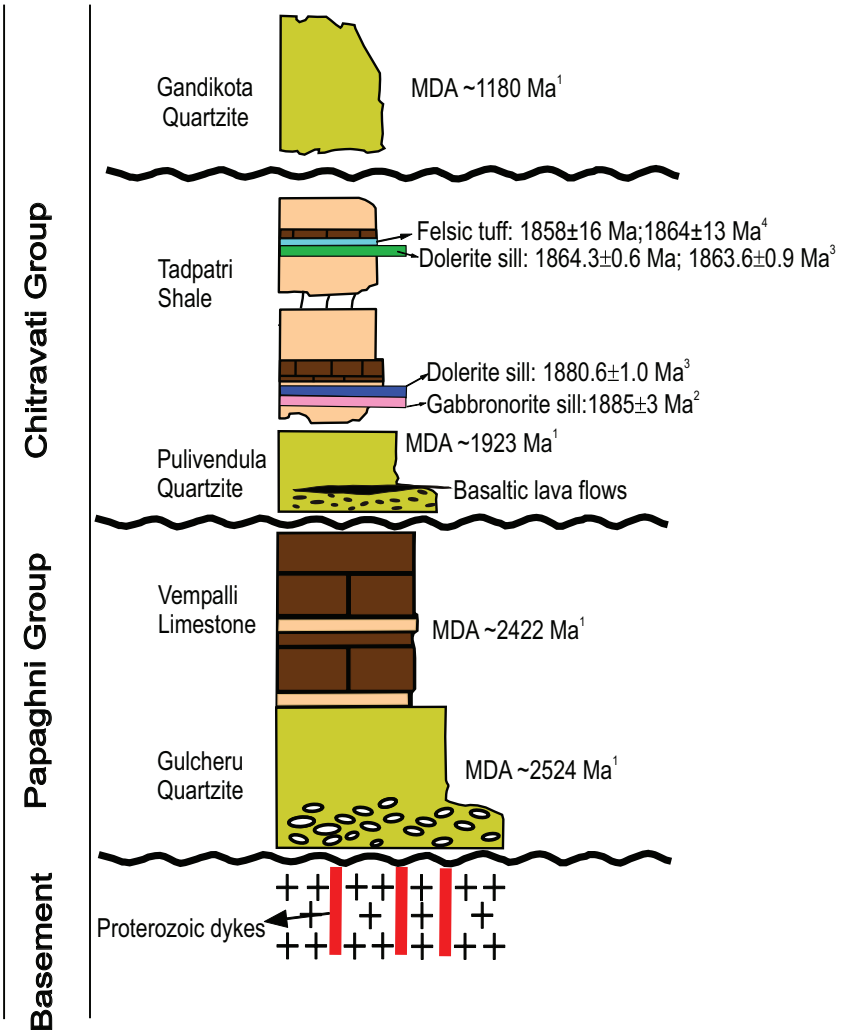


Fig. 10. Lithostratigraphy of the Papaghni and Chitravathi groups, Cuddapah basin, after Saha and Patranabis-Deb, 2014.: 1 = maximum depositional ages after Collins et al. (2015), 2 = age of sill after French et al. (2008); 3 = sill ages from Sarma et al. (2020); 4 = ages of felsic tuffs from Sheppard et al., (2017). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in recent years (e.g., Belica et al., 2014; Klein et al., 2016; Antonio et al., 2017; Liu et al., 2018). But the paleoposition of India at 1865 Ma is unknown. Various paleogeographic reconstruction models have shown India in spatial proximity to different cratons like Yilgarn, Baltica, Kalahari, Siberia, etc. (Zhao et al., 2004; Hou et al., 2008; Meert et al., 2010; Pesonen et al., 2012; Radhakrishna et al., 2013b; Belica et al., 2014; Klein et al., 2016; Antonio et al., 2017; Liu et al., 2018) based on various parameters. The most probable reason for such a situation would be the presence of limited high-quality key paleomagnetic poles during Paleoproterozoic from most cratons/continents. In this regard, the results of present study fill a vital data gap during ~1880–1860 Ma, both in the form of precise geochronology as well as high-quality paleomagnetic poles.

In this section, we propose global reconstructions at ~1885 Ma and ~ 1865 Ma based on available key paleopoles of high quality (Table 3). The reconstruction presented here is based on the meaningful geometric fitting of coeval mafic magmatic intrusions like dykes and sills. Here, we have shown the cratons/continents in a unique assemblage, to bring them in close proximity permissible within the constraints of the geology and paleomagnetic data. The reader must note that while the paleolatitude and azimuthal orientation of each craton/continent is fixed but the paleolongitude and polarity are ambiguous (Buchan, 2014). The Euler rotation parameters used are given in Table 3. The presented paleogeographic reconstruction model (Fig. 11) was prepared using the free GPLATES software (http://www.gplates.org/) has been used.

Since Laurentia has not yet attained its present configuration at ~1.89 Ga (Evans and Mitchell, 2011), the poles from its constituting blocks can be used independently. Hence, the Superior craton is treated as an independent block in present reconstruction. The Ghost dykes paleopole (of Slave craton) can represent the Slave craton paleoposition at ~1.89 Ga. Baltica craton was not assembled in its present-day configuration at 1900–1700 Ma (Bogdanova et al., 2008; Elming et al., 2010) and hence only the Baltic shield (Fennoscandia) is used here for reconstruction. According to Söderlund et al. (2010) the Zimbabwe craton and Kaapvaal craton had accreted along Limpopo Belt to form proto-Kalahari during ~2.0 Ga to 1.9 Ga. Further, available data suggests that Australia assembled in its present configuration no sooner than ~1.80 Ga (e.g., Cawood and Korsch, 2008; Li and Evans, 2011; Betts et al., 2016) and hence the Yilgarn poles only represent the western Australia craton and not the entire Australia at ~1.88 Ga and ~ 1.86 Ga. Lastly, the Southern Indian block (constituted by Dharwar, Bastar, and, Singhbhum cratons) and Northern India block (Aravalli, and Delhi-Bundelkhand cratons) were assembled at 2100–1900 Ma (Mohanty, 2010), so entire India is treated as a single block.

The position of India around the equator is constrained by the Hampi dyke swarm pole (present study) at ~1.88 Ga and Sangala Palli- Chimiryala swarm pole (present study) at ~1.86 Ga. The paleoposition of

Table 3
Paleomagnetic poles used for Paleogeographic reconstruction at ~1880 Ma and ~ 1860 Ma.

Craton/Continent	Formation	Age (Ma)	Reference	P_{lat} (°N)	P_{long} (°E)	A95 (°)	Reference	Euler pole parameters (λ (°N); Φ (°E); ω (°))
Dharwar craton/India	Hampi dyke swarm	1887–1881	Present study	22	334	3.6	Present Study	3.9; −106.8; 61.3
Dharwar craton/ India	**Sangala Palli- Chimiryala swarm**	**1867–1861**	**Present study**	**−13**	**152**	**7.8**	**Present Study**	**−0.5; 67.7; 285.2**
Yilgarn craton/ W. Australia	Boonadagin dykes	1892–1884	Stark et al., 2019	−46.8	234.9	5.9	Liu et al., 2019	−3.5; 133.6; −48.3
Superior craton/ Laurentia	Molson Dykes (B)	1884–1877	Halls and Heaman, 2000	28.9	218	3.8	Evans and Halls, 2010	32.3; 53.5; −169.4
Slave craton	Ghost Dykes	1887–1884	Buchan et al., 2016; Davis and Bleeker, 2007	2	−106	6	Buchan et al., 2016	43.8; 78.4; −170.3
Amazonia craton	Rhyolitic lava flows (SF1)	1877.4 ± 4.3	Antonio et al., 2017	−24.7	319.7	16.9	Antonio et al., 2017	46.6; 93.6; −258
Amazonia craton	**Rhyolitic dyke (SF2)**	**1853.7 ± 6.2**	**Antonio et al., 2017**	**−31.1**	**220.1**	**5.0**	**Antonio et al., 2017**	**−48.9; −178; −103.1**
Kaapval craton	Waterburg sills	1872–1879	Hanson et al., 2004	15.6	17.1	8.9	Hanson et al., 2004	31.6; −141.7; 204.6
Zimbabwe craton	Mashonaland sills	1872–1886	Sodurlund et al., 2010 Hanson et al., 2011	8	338	5	Bates and Jones, 1996	29.1; −134.3; 252.7
Baltica shield	Mean Baltica	1879–1886	Pesonen et al., 2003	41	233	5	Pesonen et al., 2003	28.2; 56.4; −173.6
Baltica shield	**Keruu dykes**	**1870–1867**	**Klein et al., 2016**	**45.7**	**230.9**	**5.5**	**Klein et al., 2016**	**24.7; 52.9; 182.3**
Siberia craton	Lower Akitkan group	1878–1866	Didenko et al., 2009 Neymark et al., 1991	−30.8	98.7	3.5	Didenko et al., 2009	−28.9; 32.1; 287.4
Siberia craton	**Upper Akitkan group**	**1863–1854**	**Didenko et al., 2009 Larin et al., 2003**	**−22.1**	**97.5**	**6.9**	**Didenko et al., 2009**	**−23.8; 27.6; −74.6**

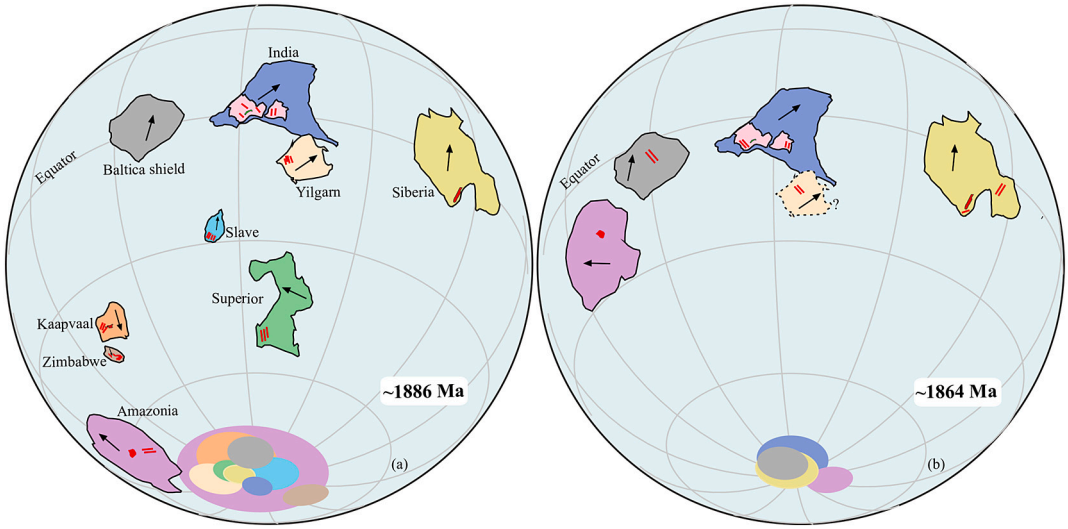


Fig. 11. Paleogeographic reconstructions (a) at ~1.88 Ga and (b) ~1.86 Ga based on key paleopoles from various cratons/continents compiled in Table 3. Respective euler rotation parameters (to the absolute framework) used in these reconstructions are given in Table 3. The pole and craton/continent pairs are depicted by same colour combination. The mafic dykes and related coeval magmatism in all cratons are shown by red colour. The black arrows demonstrate the present day North for individual cratons/continents. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Western Australia craton is near the equator, as determined by the Boonadgin dykes paleopole (Liu et al., 2018). A geospatial connection between South India block and Western Australia craton was proposed by Mohanty (2010, 2012), based on similar tectonic histories of coeval orogenies viz., the Capricorn orogeny of Western Australia and the Central India Tectonic Zone of India. Further, the possible genetic linkage of ~1.89 Ga −1.88 Ga Boonadgin dyke swarm with the ~1.88 Ga Bastar-Cuddapah LIP is supported by many recent publications (Liu et al., 2018; Shellnutt et al., 2018, 2019; Stark et al., 2017. In the present reconstruction (Fig. 11) the western margin of Western Australia craton is placed in vicinity to the eastern margin of India, both proximal to the equator, at ~1.88 Ga. The paleoposition of Superior craton and Slave cratons are constrained at moderate paleolatitudes by Molson dyke (Halls and Heaman, 2000; Evans and Halls, 2010) and Ghost dyke (Buchan et al. (2016) paleopoles, respectively. Hence the Superior and Slave cratons can be placed at moderate latitude next to the southern margin of Western Australia craton.

Paleogeographic configuration at ~1.89 Ga (Fig. 11a) presents, the occurrence of mafic dykes of EDC (Hampi dyke swarm), Bastar craton (NW-SE) and Yilgarn craton (Boonadagin dykes), in a radial fashion with a common focus, which could serve as a plume source for the generation of these mafic intrusions in respective cratons. The approximate plume centre could be situated in the vicinity of the eastern margin of India. Moreover, based on matching radial pattern of dykes from Slave craton (Ghost dykes) and superior craton (Molson dykes), it can be inferred that these dykes perhaps have originated from coeval but different source,

owing to the large geospatial distance. The proximity of plume centre to the eastern margin of India is supported by the presence of widespread and extensive volcanism at 1.88–1.86 Ga in the form of mafic dykes and sills across Bastar and Dharwar cratons. The sub-horizontal to horizontal magma flow within ~1.88 Ga dykes, based on AMS study (Ernst and Baragar, 1992), indicates that Western Australia lied away from the plume centre. Hence, it can be argued that the ~1.88 Ga mantle plume, centred proximal to the eastern margin of India, had emplaced radial dyke swarm across Dharwar (India), Western Australia. Superior, and Slave cratons have witnessed emplacement of coeval dykes from a different source.

Similarly, Fig. 11b shows, the emplacement of mafic magmatism in the Dharwar craton (Sangala Palli-Chimiryala swarm), Bastar craton (Sonakhan dykes) and Yilgarn craton (Yalgoo dykes) also form a radial pattern converging towards the focal point of plume source, which could be responsible for ~1.89 Ga magmatism. Hence, we presume that eminent of two distinguishable mafic episodes (1.89 Ga and 1.86 Ga) were emplaced from a single plume source within a time span of <25 Ma. At 1.86 Ga, we placed Yilgarn craton at low latitudes as assume of low drift rate similar to India. However, key Paleopole of Yalgoo dykes should be needed to ascertain the valid paleolatitudinal position of Yilgarn craton and also address the spatial linkage with neighbouring continents at that time. We are not able to extend the paleogeographic positions for other cratons due to ambiguity of ages and paleopoles at ~1.86 Ga.

Key plaeopoles constrain Baltic shield (Fennoscandia) paleoposition at ~1.88 Ga and 1.86 Ga at the equator (Pesonen et al., 2003; Klein et al., 2016). The spatial link between Fennoscandia and India during Paleoproterozoic to Mesoproterozoic (for ~370 Ma, i.e., ~1.77 Ga to ~1.40 Ga) is supported by paleomagnetic, geological, tectonic and metallogenic evidence (Pisarevsky et al., 2013; Shankar et al., 2018). And hence we have placed the Fennoscandia in the vicinity of the south-western margin of India at ~1.88 Ga and ~ 1.86 Ga, to allow it to accrete with India as mentioned earlier.

In present reconstruction, the paleoposition of proto-Kalahari lies at moderate latitudes at ~1.88 Ga, as permitted by Mashonaland and Waterberg sills' poles (Hanson et al., 2004; Bates and Jones, 1996).

Siberia is kept at the equatorial position and is constrained by Lower Akitan group pole at ~1.88 Ga (Didenko et al., 2009) and by Upper Akitan group pole at ~1.87 Ga (Didenko et al., 2009), indicating minimal latitudinal movement of Siberia during this time interval. The Amazonia craton paleoposition is placed near pole at ~1.88 Ga by Rhyolitic lava flows (SF1) pole (Antonio et al., 2017) and near the equator at ~1.86 Ga by Rhyolitic dyke (SF2) poles (Antonio et al., 2017). Antonio et al., 2017 had attributed such rapid change is the position of Amazonia's paleopole in short time span of ~20 Ma to the True polar wander.

The relative changes in paleopositions during ~1.88 Ga to ~1.86 Ga can be commented upon for India, Baltica, Siberia and Amazonia. This is mainly due to lack of paleomagnetic data at ~1.86 Ga for another craton/continents. However, India, Baltica, and Siberia had experienced about 10° latitudinal drift at a rate of ~5.55 cm/yr during ~1.88–1.86 Ga (~ 20 Ma) towards the south. On the contrary, the Amazonia had a rapid northward shift 45° or ~ 24.9 cm/yr, during ~1.88–1.86 Ga (Antonio et al., 2017). Further, Fig. 11 also shows, independently drifting of cratonic/continental blocks during ~1.88–1.86 Ga, and do not support the amalgamation of Columbia or Nuna supercontinent in the early Paleoproterozoic time, as also suggested by Liu et al. (2018).

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7. Conclusions

1. We are reporting an age of 1867.1 ± 1.0 Ma (MSWD = 1.02), on coarse-grained dolerite sample (TP 19), collected from N77°E trending dyke near Ontillu village, thereby confirming the presence of ~1864 Ma mafic magmatic episode in the EDC with a large spatial extent of about ~400 km within the brief period of ~5 Ma.
2. The mean paleomagnetic result for ~1864 Ma event from 12 sites yields Declination =107°; Inclination = 24°, with k = 21; $\alpha_{95} = 10°$. The corresponding paleopole situated at 11°S, 152°E (dp = 6°, dm = 10°). This is the first key plaeopole at ~1864 Ma reported from Dharwar craton.
3. The ~1.89 Ga key paleopole is updated based on results of calculated ChRM directions Declination = 294°, Inclination = −12°, with k = 11 and $\alpha_{95} = 5°$. The corresponding paleopole estimated to be located at 21°N, 336°E (dp = 3°, dm = 5°).
4. The paleogeographic reconstruction presented here, constrained by paleomagnetic data and supported by the meaningful geometric fitting of coeval mafic events like dykes and sills, reveals the presence of radial dyke emplacement across India, Western Australia, Superior, and Slave at ~1.89 Ga.
5. During the interval of ~1.89–1.86 Ga, India, Baltica and Siberia moved southwards at the rate of ~5.55 cm/yr. The Amazonia craton has moved in the opposite direction at a much rapid rate of ~24.9 cm/yr during the same time.

Author statement

V. Parashuramulu: Carried out the field work; geochronology, paleomagnetic work and Data interpretation and initial and final draft preparation.

Ravi Shankar: Data interpretation and initial and final draft preparation.

D. Srinivasa Sarma: Geochronology, data interpretation and initial and final draft preparation.

E. Nagaraju: Contributed in field work; data interpretation and write-up of initial and final version.

N. Ramesh Babu: Contributed in field work; data interpretation and write-up of initial and final version.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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